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(54) **TREATMENT OF NEUROFIBROMATOSIS
TYPE 1 (NF1) ASSOCIATED PLEXIFORM
NEUROFIBROMAS (PN) IN PEDIATRIC
PATIENTS WITH MIRDAMETINIB**

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(71) Applicant: **SpringWorks Therapeutics, Inc.**,
Stamford, CT (US)

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(72) Inventors: **Uchenna H. ILOEJE**, Stamford, CT
(US); **Abraham J. LANGSETH**,
Stamford, CT (US); **Todd Shearer**,
Stamford, CT (US)

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(57)

ABSTRACT

The present disclosure relates to methods for treating plexi-
form neurofibromas associated with neurofibromatosis type
1 (NF1-PN), comprising administering to a pediatric patient
(e.g., 2 to 15 years of age) in need thereof mirdametinib or
a pharmaceutically acceptable salt thereof.

**TREATMENT OF NEUROFIBROMATOSIS
TYPE 1 (NF1) ASSOCIATED PLEXIFORM
NEUROFIBROMAS (PN) IN PEDIATRIC
PATIENTS WITH MIRDAMETINIB**

[0001] The present application is a continuation of U.S. patent application Ser. No. 18/648,426, filed Apr. 28, 2024, which is a continuation of U.S. patent application Ser. No. 18/357,988, filed Jul. 24, 2023, now U.S. Pat. No. 12,011,424, which is a continuation of U.S. patent application Ser. No. 18/185,080, filed Mar. 16, 2023, now U.S. Pat. No. 11,819,487, which claims the benefit of U.S. Provisional Application Nos. 63/321,036, filed Mar. 17, 2022, and 63/321,046, filed Mar. 17, 2022, each of which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present disclosure relates to methods for treating a pediatric patient (e.g., 2 to 15 years of age) who has neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) by orally administering an effective amount of mirdametininb to the patient.

BACKGROUND

[0003] Mirdametininb is an allosteric, small molecule targeting mitogen-activated protein kinase kinase (MEK).

[0004] Weiss describes a Phase II clinical trial of mirdametininb in subjects at least 16 years of age with neurofibromatosis type 1 who have a plexiform neurofibroma (Weiss et al., *J. Clin. Oncol.*, 29, 797-806, 2021).

[0005] Prior to the present invention, the safety and efficacy of mirdametininb in pediatric patients 15 years of age and younger having neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN), such as those with progressive PN, was unknown.

[0006] However, because objective responses are rarely complete and disease recurrence after completion of therapy is common in various tumors and cancers, there is a need to find a better treatment.

BRIEF SUMMARY OF THE INVENTION

[0007] One aspect of the present invention is a method of treating a pediatric human patient (e.g., 2 to 15 years of age) who has neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) by orally administering an effective amount of mirdametininb to the patient. In one embodiment, the patient has symptomatic, inoperable plexiform neurofibromas.

[0008] Another aspect is a method of treating a pediatric human patient 2 to 15 years of age who has neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) that is progressing or causing significant morbidity comprising orally administering an effective amount of mirdametininb to the patient.

[0009] In one embodiment of any of the methods described herein, the patient has progressive PN (i.e., a 20% increase in PN volume documented by comparison of two MRI scans in the time period of 12 months or less prior to the first dose of mirdametininb).

[0010] In one embodiment of any of the methods described herein, the patient has PNs that cause significant morbidity.

[0011] In one embodiment of any of the methods described herein, the patient has head and neck lesions that are compromising the airway or great vessels, brachial or lumbar plexus lesions that are causing nerve compression and loss of function, lesions causing major deformity or are significantly disfiguring, lesions of the extremity that cause limb hypertrophy or loss of function, or painful lesions. In one embodiment, the lesions causing major deformity or are significantly disfiguring are tumors of the head and neck or those on other areas of the body that are unable to be concealed by standard garments.

[0012] In one embodiment of any of the methods described herein, the patient has paraspinal lesions.

[0013] In one embodiment of any of the methods described herein, the patient has a Lansky performance of at least 60%.

[0014] In one embodiment of any of the methods described herein, the patient has the clinical diagnosis of NF1 using the NIH Consensus Conference and one or more of the following:

[0015] (a) six or more café-au-lait macules with a diameter >5 mm in prepubertal and >15 mm in post-pubertal individuals;

[0016] (b) freckling in axilla or inguinal regions;

[0017] (c) optic glioma;

[0018] (d) two or more Lisch nodules;

[0019] (e) a distinctive bony lesion (dysplasia of the sphenoid bone or dysplasia of thinning of long bone cortex); and

[0020] (f) a first degree relative with NF1.

[0021] In one embodiment of any of the methods described herein, the patient has a constitutional NF1 mutation documented in a Clinical Laboratory Improvement Amendments/College of American Pathologists certified lab.

[0022] In one embodiment of any of the methods described herein, the patient either (a) has a parent diagnosed with NF1 and one or more criteria of (1) through (7), or (b) does not have a parent diagnosed with NF1 but has two or more criteria of (1) through (7):

[0023] (1) six or more café-au-lait macules over 5 mm in greatest diameter in prepubertal individuals and over 15 mm in greatest diameter in post-pubertal individuals;

[0024] (2) freckling in the axillary or inguinal region;

[0025] (3) two or more neurofibromas of any type or one plexiform neurofibroma;

[0026] (4) optic pathway glioma;

[0027] (5) two or more iris Lisch nodules identified by slit lamp examination or two or more choroidal abnormalities (defined as bright, patchy nodules imaged by optical coherence tomography (OCT)/near-infrared reflectance (NIR) imaging);

[0028] (6) a distinctive osseous lesion (such as sphenoid dysplasia, anterolateral bowing of the tibia, or pseudarthrosis of a long bone); and

[0029] (7) a heterozygous pathogenic NF1 variant with a variant allele fraction of 50% in apparently normal tissue such as white blood cells.

[0030] In one embodiment of any of the methods described herein,

[0031] (a) for a patient having a body surface area no more than 0.69 m^2 , the patient is initially orally administered 1 mg mirdametinib twice daily (i.e., a total of 2 mg daily),

[0032] (b) for a patient having a body surface area of 0.7 to 1.04 m^2 , the patient is initially orally administered 2 mg mirdametinib twice daily (i.e., a total of 4 mg daily),

[0033] (c) for a patient having a body surface area of 1.05 to 1.49 m^2 , the patient is initially orally administered 3 mg mirdametinib twice daily (i.e., a total of 6 mg daily), and

[0034] (d) for a patient having a body surface area of at least 1.5 m^2 , the patient is initially orally administered 4 mg mirdametinib twice daily (i.e., a total of 8 mg daily).

[0035] The present inventors discovered that a lower dose than 2 mg/m^2 body surface area is effective in pediatric patients (such as those having a body surface area no more than 0.69 m^2) despite tumors often being more aggressive in such patients compared to adult patient. In one embodiment, for a pediatric patient having a body surface area no more than 0.69 m^2 , the patient is initially orally administered 1 mg mirdametinib twice daily (i.e., a total of 2 mg daily).

[0036] In another embodiment, for a patient 2 to 10 years old and having a body surface area no more than 0.69 m^2 , the patient is initially orally administered 1 mg mirdametinib twice daily (i.e., a total of 2 mg daily). In yet another embodiment, for a patient 2 to 8 years old and having a body surface area no more than 0.69 m^2 , the patient is initially orally administered 1 mg mirdametinib twice daily (i.e., a total of 2 mg daily). In yet another embodiment, for a patient 2 to 7 years old and having a body surface area no more than 0.69 m^2 , the patient is initially orally administered 1 mg mirdametinib twice daily (i.e., a total of 2 mg daily). In yet another embodiment, for a patient 2 to 6 years old and having a body surface area no more than 0.69 m^2 , the patient is initially orally administered 1 mg mirdametinib twice daily (i.e., a total of 2 mg daily). In yet another embodiment, for a patient 2 to 5 years old and having a body surface area no more than 0.69 m^2 , the patient is initially orally administered 1 mg mirdametinib twice daily (i.e., a total of 2 mg daily).

[0037] In another embodiment of any of the methods described herein, the method comprises orally administering 1 mg mirdametinib twice daily (i.e., a total of 2 mg daily).

[0038] In another embodiment of any of the methods described herein, the maximum oral daily dose administered to the patient is 4 mg mirdametinib twice daily (i.e., a total of 8 mg daily).

[0039] In one embodiment of any of the methods described herein, about 2 mg/m^2 mirdametinib is orally administered to the patient twice daily.

[0040] In one embodiment of any of the methods described herein, over each four week period, the mirdametinib is administered for the first three weeks and discontinued for the last one week.

[0041] In one embodiment of any of the methods described herein, the patient has at least a 20% reduction in plexiform neurofibroma volume as determined by volumetric magnetic resonance imaging analysis following treatment with mirdametinib.

[0042] In one embodiment of any of the methods described herein, the treatment results in decreased pain intensity.

[0043] In one embodiment of any of the methods described herein, the treatment results in decreased pain interference.

[0044] In one embodiment of any of the methods described herein, the dose administered is reduced due to an adverse event, wherein the dose is reduced as follows:

[0045] (a) if the oral dose at the time of the event is 1 mg mirdametinib twice daily, then the reduced daily dose is 1 mg orally administered in the morning only;

[0046] (b) if the oral dose at the time of the event is 2 mg mirdametinib twice daily, then the reduced daily dose is 2 mg orally administered in the morning and 1 mg orally administered in the afternoon or evening;

[0047] (c) if the oral dose at the time of the event is 3 mg mirdametinib twice daily, then the reduced daily dose is 2 mg orally administered twice daily; and

[0048] (d) if the oral dose at the time of the event is 4 mg mirdametinib twice daily, then the reduced daily dose is 3 mg orally administered twice daily.

[0049] In one embodiment of any of the methods described herein, the adverse event resulting in the dose reduction is acneiform.

[0050] In one embodiment of any of the methods described herein, the method further comprises prior to treatment (i) determining whether to select mirdametinib as a treatment for the patient, and (ii) selecting mirdametinib as a treatment for the patient at least partially based on its objective response rate, where the objective response rate is defined as at least a 20% decrease in tumor size using centrally read MRI volumetric analysis. In one embodiment, in step (i), mirdametinib is selected based on a response rate of at least 70%. In another embodiment, in step (i), mirdametinib is selected based on a response rate of at least 75%. In yet another embodiment, in step (i), mirdametinib is selected based on a response rate of at least 80%. In yet another embodiment, in step (i), mirdametinib is selected based on a response rate of at least 85%. In yet another embodiment, in step (i), mirdametinib is selected based on a response rate of at least 90%. In yet another embodiment, in step (i), mirdametinib is selected based on a response rate of at least 95%.

[0051] Another aspect is a method of treating a pediatric patient 2 to 5 years of age who has neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) by orally administering 1 mg mirdametinib twice daily to the patient.

[0052] Yet another aspect of the present invention is a method of treating a pediatric patient 2 to 6 years of age who has neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) by orally administering 1 mg mirdametinib twice daily to the patient

[0053] Yet another aspect of the present invention is a method of treating a pediatric patient 2 to 7 years of age who has neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) by orally administering 1 mg mirdametinib twice daily to the patient

[0054] Yet another aspect of the present invention is a method of treating a pediatric patient 2 to 8 years of age who has neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) by orally administering 1 mg mirdametinib twice daily to the patient.

[0055] Yet another aspect of the present invention is a method of treating a pediatric patient 2 to 10 years of age who has neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) by orally administering 1 mg mirdametinib twice daily to the patient.

[0056] In one embodiment of any of the methods described herein, the pediatric patient has had no prior exposure to MEK inhibitors.

[0057] In some aspects, the mirdametinib is dispersible in a potable liquid or orodispersible in a patient's saliva. In some aspects, the mirdametinib is administered orally as a solid dosage form. In some aspects, the solid dosage form is a tablet or capsule. In some aspects, the solid dosage form is a capsule. In some aspects, the solid dosage form is a dispersible tablet.

[0058] In some aspects, the mirdametinib is administered as a monotherapy.

DETAILED DESCRIPTION OF THE INVENTION

I. Definitions

[0059] To facilitate understanding of the disclosure set forth herein, a number of terms are defined below.

[0060] Generally, the nomenclature used herein and the laboratory procedures in organic chemistry, medicinal chemistry, and pharmacology described herein are those well-known and commonly employed in the art. Unless defined otherwise, all technical and scientific terms used herein generally have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs.

[0061] In this specification and the appended claims, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. The terms “a” (or “an”), as well as the terms “one or more,” and “at least one” can be used interchangeably herein. In certain aspects, the term “a” or “an” means “single.” In other aspects, the term “a” or “an” includes “two or more” or “multiple.”

[0062] Furthermore, “and/or” where used herein is to be taken as specific disclosure of each of the two specified features or components with or without the other. Thus, the term “and/or” as used in a phrase such as “A and/or B” herein is intended to include “A and B,” “A or B,” “A” (alone), and “B” (alone). Likewise, the term “and/or” as used in a phrase such as “A, B, and/or C” is intended to encompass each of the following aspects: A, B, and C; A, B, or C; A or C; A or B; B or C; A and C; A and B; B and C; A (alone); B (alone); and C (alone).

[0063] The term “mirdametinib” refers to the single enantiomer N—((R)-2,3-dihydroxypropoxy)-3,4-difluoro-2-(2-fluoro-4-iodo-phenylamino)benzamide (free base form). The teachings throughout the specification regarding mirdametinib equally apply to pharmaceutically acceptable salts of mirdametinib. For instance, the disclosure of a method of treating neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) with mirdametinib also means that a pharmaceutically acceptable salt of mirdametinib can be administered to treat NF1 associated inoperable PN.

[0064] The term “mg/m²” refers to the dose in milligrams per m² body surface area of the patient.

[0065] The term “patient” or “subject” refers to a human.

[0066] The term “pediatric” as used herein, refers to a human subject 2 to 15 years of age at the time of treatment. The term “pediatric” can be further divided into various subpopulations including: neonates (from birth through the first 28 days of life); infants (29 days of age to less than two years of age); children (two years of age to less than 12 years of age); and adolescents (12 years of age through 15 years of age. See, e.g., Berhman R E, Kliegman R, Arvin A M, Nelson W E. *Nelson Textbook of Pediatrics*, 15th Ed. Philadelphia: W.B. Saunders Company, 1996; Rudolph A M, et al. *Rudolph's Pediatrics*, 21st Ed. New York: McGraw-Hill, 2002; and Avery M D, First L R. *Pediatric Medicine*, 2nd Ed. Baltimore: Williams & Wilkins; 1994. In some additional embodiments, a pediatric patient is 2 to 8 years of age, 2 to 7 years of age, 2 to 6 years of age or 2 to 5 years of age

[0067] The term “dispersible” as used herein refers to a composition (e.g., a tablet, powder, granules, minitabets, or pellets) which disintegrates and/or dissolves when combined with water or another potable liquid (e.g., a non-water beverage), or a subject's own saliva when placed in the subject's mouth, with or without the addition of agitation or temperature modification. In some aspects, the dispersible composition disintegrates or dissolves within 10 minutes, 9 minutes, 8 minutes, 7 minutes, 6 minutes, 5 minutes, 4 minutes, 3 minutes, 2 minutes, or 1 minute after being combined with water or another potable liquid. Such disintegration or dissolution need not be complete. For example, a dispersible tablet may dissolve almost entirely, but some undissolved particulate matter may remain. Dispersible formulations of mirdametinib suitable for the methods described herein include those described in U.S. Pat. No. 11,571,402, which is hereby incorporated by reference.

[0068] The term “orodispersible” refers to a composition which is capable of dissolving or disintegrating in a subject's mouth (i.e., dissolving or disintegrating in a subject's saliva) if administered orally, without a requirement of first dissolving or disintegrating in a separate container.

[0069] As used herein, the terms “treat,” “treated,” and “treating” mean both therapeutic treatment and prophylactic or preventative measures wherein the object is to prevent or slow down (lessen) an undesired physiological condition, disorder, or disease, or obtain beneficial or desired clinical results. Thus, those in need of treatment include those already diagnosed with or suspected of having the disorder. Beneficial or desired clinical results include, but are not limited to, alleviation of symptoms; diminishment of the extent of a condition, disorder, or disease; stabilized (i.e., not worsening) state of condition, disorder, or disease; delay in onset or slowing of condition, disorder, or disease progression; amelioration of the condition, disorder, or disease state or remission (whether partial or total), whether detectable or undetectable; an amelioration of at least one measurable physical parameter, not necessarily discernible by the patient; or enhancement or improvement of condition, disorder, or disease. Treatment includes eliciting a clinically significant response without excessive levels of side effects. Treatment also includes prolonging survival as compared to expected survival if not receiving treatment. The term “therapeutically effective amount” is meant to include the amount of a compound that, when administered, is sufficient to prevent development of, or alleviate to some extent, one or more of the symptoms of a disorder, disease, or condition being treated. The term “therapeutically effective amount”

also refers to the amount of a compound that is sufficient to elicit the biological or medical response of a cell, tissue, system, animal, or human, which is being sought by a researcher, veterinarian, medical doctor, or clinician.

[0070] In certain aspects, a subject is successfully “treated” for a tumor, according to the methods described herein if the patient shows one or more of the following: a reduction in the size of the tumor; relief of one or more symptoms associated with the specific tumor; a reduction in the volume of the tumor; improvement in quality of life; increased progression-free survival (PFS), disease-free survival (DFS), overall survival (OS), metastasis-free survival (MFS), complete response (CR), minimal residual disease (MRD), partial response (PR), stable disease (SD), a decrease in progressive disease (PD), an increased time to progression (TTP), or any combination thereof. In some aspects, nationally or internationally accepted standards of treatment outcomes in a given tumor can be used to determine whether an effective amount of mirdametinib meets any of these particular endpoints (e.g., CR, PFS, PR).

[0071] In certain aspects, a subject is successfully “treated” for cancer, e.g., ovarian cancer, according to the methods described herein if the patient shows one or more of the following: a reduction in the number of or complete absence of cancer cells; relief of one or more symptoms associated with the specific cancer; reduced morbidity and mortality; improvement in quality of life; increased progression-free survival (PFS), disease-free survival (DFS), overall survival (OS), metastasis-free survival (MFS), complete response (CR), minimal residual disease (MRD), partial response (PR), stable disease (SD), a decrease in progressive disease (PD), an increased time to progression (TTP), or any combination thereof. In some aspects, nationally or internationally accepted standards of treatment outcomes in a given cancer can be used to determine whether an effective amount of mirdametinib meets any of these particular endpoints (e.g., CR, PFS, PR).

[0072] The terms “pharmaceutically acceptable carrier,” “pharmaceutically acceptable excipient,” “physiologically acceptable carrier,” or “physiologically acceptable excipient” refer to a pharmaceutically acceptable material, composition, or vehicle, such as a liquid or solid filler, diluent, excipient, solvent, or encapsulating material. In one embodiment, each component is “pharmaceutically acceptable” in the sense of being compatible with the other ingredients of a pharmaceutical formulation, and suitable for use in contact with the tissue or organ of humans and animals without excessive toxicity, irritation, allergic response, immunogenicity, or other problems or complications, commensurate with a reasonable benefit/risk ratio. See *Remington: The Science and Practice of Pharmacy*, 21st Edition, Lippincott Williams & Wilkins: Philadelphia, PA, 2005; *Handbook of Pharmaceutical Excipients*, 5th Edition, Rowe et al., Eds., The Pharmaceutical Press and the American Pharmaceutical Association: 2005; and *Handbook of Pharmaceutical Additives*, 3rd Edition, Ash and Ash Eds., Gower Publishing Company: 2007; *Pharmaceutical Preformulation and Formulation*, Gibson Ed., CRC Press LLC: Boca Raton, FL, 2004 (incorporated herein by reference).

[0073] The term “pharmaceutically acceptable salts” refers to the relatively non-toxic, inorganic and organic acid addition salts of mirdametinib. These salts can be prepared in situ in the administration vehicle or the dosage form manufacturing process, or by separately reacting mirdame-

tinib in its free base form with a suitable organic or inorganic acid, and isolating the salt thus formed during subsequent purification. Representative salts include, but are not limited to, hydrobromide, hydrochloride, sulfate, bisulfate, phosphate, nitrate, acetate, valerate, oleate, palmitate, stearate, laurate, benzoate, lactate, phosphate, tosylate, citrate, maleate, fumarate, succinate, tartrate, naphthylate, mesylate, glucoheptonate, lactobionate, and laurylsulphonate salts. See, e.g., Berge et al. (1977) “Pharmaceutical Salts”, *J. Pharm. Sci.* 66:1-19.

[0074] The pharmaceutically acceptable salts of mirdametinib also include the conventional nontoxic salts or quaternary ammonium salts of the compounds, e.g., from non-toxic organic or inorganic acids. For example, such conventional nontoxic salts include those derived from inorganic acids such as hydrochloride, hydrobromic, sulfuric, sulfamic, phosphoric, nitric, and the like; and the salts prepared from organic acids such as acetic, propionic, succinic, glycolic, stearic, lactic, malic, tartaric, citric, ascorbic, palmitic, maleic, hydroxymaleic, phenylacetic, glutamic, benzoic, salicylic, sulfanilic, 2-acetoxybenzoic, fumaric, toluenesulfonic, methanesulfonic, ethane disulfonic, oxalic, and isothionic.

[0075] The term “about” or “approximately” means an acceptable error for a particular value as determined by one of ordinary skill in the art, which depends in part on how the value is measured or determined. In certain embodiments, the term “about” or “approximately” means within 1, 2, 3, or 4 standard deviations. In certain embodiments, the term “about” or “approximately” means within 50%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, or 0.05% of a given value or range.

[0076] Unless the context requires otherwise, the terms “comprise,” “comprises,” and “comprising” are used on the basis and clear understanding that they are to be interpreted inclusively, rather than exclusively, and that Applicant intends each of those words to be so interpreted in construing this patent, including the claims below.

II. Methods of Treatment

[0077] Methods for treating plexiform neurofibromas associated with neurofibromatosis type 1 (NF1-PN) comprising administering to a pediatric patient (e.g., 2 to 15 years of age) in need thereof mirdametinib or a pharmaceutically acceptable salt thereof are provided herein.

[0078] In one embodiment of any of the methods described herein, the patient has progressive PN (i.e., a 20% increase in PN volume documented by comparison of two MRI scans in the time period of 12 months or less prior to the first dose of mirdametinib).

[0079] In one embodiment of any of the methods described herein, the patient has PNs that cause significant morbidity.

[0080] In one embodiment of any of the methods described herein, the patient has head and neck lesions that are compromising the airway or great vessels, brachial or lumbar plexus lesions that are causing nerve compression and loss of function, lesions causing major deformity or are significantly disfiguring, lesions of the extremity that cause limb hypertrophy or loss of function, or painful lesions. In one embodiment, the lesions causing major deformity or are significantly disfiguring are tumors of the head and neck or those on other areas of the body that are unable to be concealed by standard garments.

[0081] In one embodiment of any of the methods described herein, the patient has paraspinous lesions.

[0082] In one embodiment of any of the methods described herein, the patient has a Lansky performance of at least 60%.

[0083] In one embodiment of any of the methods described herein, the patient has the clinical diagnosis of NF1 using the NIH Consensus Conference.

[0084] In one embodiment of any of the methods described herein, the patient has the clinical diagnosis of NF1 using the NIH Consensus Conference and one or more of the following:

[0085] (a) six or more café-au-lait macules with a diameter >5 mm in prepubertal and >15 mm in post-pubertal individuals;

[0086] (b) freckling in axilla or inguinal regions;

[0087] (c) optic glioma;

[0088] (d) two or more Lisch nodules;

[0089] (e) a distinctive bony lesion (dysplasia of the sphenoid bone or dysplasia of thinning of long bone cortex); and

[0090] (f) a first degree relative with NF1.

[0091] In one embodiment of any of the methods described herein, the patient has a constitutional NF1 mutation documented in a Clinical Laboratory Improvement Amendments/College of American Pathologists certified lab.

[0092] In one embodiment of any of the methods described herein, the patient is diagnosed with NF1 using the criteria for NF1 provided in Legius et al., *Genetics in Medicine*, 23, 1506-1513, 2021 (Table 1), which is hereby incorporated by reference.

[0093] In one embodiment of any of the methods described herein, the patient either (a) has a parent diagnosed with NF1 and one or more criteria of (1) through (7), or (b) does not have a parent diagnosed with NF1 but has two or more criteria of (1) through (7) (see, e.g., Legius et al., *Genetics in Medicine*, 23, 1506-1513, 2021 (Table 1)):

[0094] (1) six or more café-au-lait macules over 5 mm in greatest diameter in prepubertal individuals and over 15 mm in greatest diameter in post-pubertal individuals;

[0095] (2) freckling in the axillary or inguinal region (in one embodiment, if only café-au-lait macules and freckling are present, at least one of the two pigmentary findings (café-au-lait macules or freckling) is bilateral for this criteria to be met);

[0096] (3) two or more neurofibromas of any type or one plexiform neurofibroma;

[0097] (4) optic pathway glioma;

[0098] (5) two or more iris Lisch nodules identified by slit lamp examination or two or more choroidal abnormalities (defined as bright, patchy nodules imaged by optical coherence tomography (OCT)/near-infrared reflectance (NIR) imaging);

[0099] (6) a distinctive osseous lesion (such as sphenoid dysplasia, anterolateral bowing of the tibia, or pseudarthrosis of a long bone); and

[0100] (7) a heterozygous pathogenic NF1 variant with a variant allele fraction of 50% in apparently normal tissue such as white blood cells.

[0101] In some embodiments of any of the methods described herein, a therapeutically effective amount of mirdametinib, or a pharmaceutically acceptable salt thereof, is administered.

[0102] In some embodiments of any of the methods described herein, the mirdametinib, or a pharmaceutically acceptable salt thereof, is administered in an amount of about 1 mg/m² to about 10 mg/m² per day based on mirdametinib free base, about 1.5 mg/m² to about 9.5 mg/m² per day based on mirdametinib free base, about 2 mg/m² to about 9 mg/m² per day based on mirdametinib free base, about 2.5 mg/m² to about 8.5 mg/m² per day based on mirdametinib free base, about 3 mg/m² to about 8 mg/m² per day based on mirdametinib free base, about 3.5 mg/m² to about 7.5 mg/m² per day based on mirdametinib free base, about 4 mg/m² to about 7 mg/m² per day based on mirdametinib free base, about 4.5 mg/m² to about 6.5 mg/m² per day based on mirdametinib free base, or about 5 mg/m² to about 6 mg/m² per day based on mirdametinib free base. In some embodiments of any of the methods described herein, the mirdametinib, or a pharmaceutically acceptable salt thereof, is administered in an amount of about 1 mg/m² per day based on mirdametinib free base, about 1.5 mg/m² per day based on mirdametinib free base, about 2 mg/m² per day based on mirdametinib free base, about 2.5 mg/m² per day based on mirdametinib free base, about 3 mg/m² per day based on mirdametinib free base, about 3.5 mg/m² per day based on mirdametinib free base, about 4 mg/m² per day based on mirdametinib free base, about 4.5 mg/m² per day based on mirdametinib free base, about 5 mg/m² per day based on mirdametinib free base, about 5.5 mg/m² per day based on mirdametinib free base, about 6 mg/m² per day based on mirdametinib free base, about 6.5 mg/m² per day based on mirdametinib free base, about 7 mg/m² per day based on mirdametinib free base, about 7.5 mg/m² per day based on mirdametinib free base, about 8 mg/m² per day based on mirdametinib free base, about 8.5 mg/m² per day based on mirdametinib free base, about 9 mg/m² per day based on mirdametinib free base, about 9.5 mg/m² per day based on mirdametinib free base, or about 10 mg/m² per day based on mirdametinib free base.

[0103] In some embodiments of any of the methods described herein, the mirdametinib, or a pharmaceutically acceptable salt thereof, is administered in an amount of about 1 mg to about 10 mg per day based on mirdametinib free base, about 1.5 mg to about 9.5 mg per day based on mirdametinib free base, about 2 mg to about 9 mg per day based on mirdametinib free base, about 2.5 mg to about 8.5 mg per day based on mirdametinib free base, about 3 mg to about 8 mg per day based on mirdametinib free base, about 3.5 mg to about 7.5 mg per day based on mirdametinib free base, about 4 mg to about 7 mg per day based on mirdametinib free base, about 4.5 mg to about 6.5 mg per day based on mirdametinib free base, or about 5 mg to about 6 mg per day based on mirdametinib free base. In some embodiments of any of the methods described herein, the mirdametinib, or a pharmaceutically acceptable salt thereof, is administered in an amount of about 1 mg per day based on mirdametinib free base, about 1.5 mg per day based on mirdametinib free base, about 2 mg per day based on mirdametinib free base, about 2.5 mg per day based on mirdametinib free base, about 3 mg per day based on mirdametinib free base, about 3.5 mg per day based on mirdametinib free base, about 4 mg per day based on mirdametinib free base, about 4.5 mg per day based on mirdametinib free base, about 5 mg per day based on mirdametinib free base, about 5.5 mg per day based on mirdametinib free base, about 6 mg per day based on mirdametinib free base, about 6.5 mg per day based on mirdametinib free base, about 7 mg per day based on mirdametinib free base, about 7.5 mg per day based on mirdametinib free base, about 8 mg per day based on mirdametinib free base, about 8.5 mg per day based on mirdametinib free base, about 9 mg per day based on mirdametinib free base, about 9.5 mg per day based on mirdametinib free base, or about 10 mg per day based on mirdametinib free base.

tinib free base, about 8.5 mg/m² based on mirdametinib free base, about 9 mg/m² based on mirdametinib free base, about 9.5 mg/m² based on mirdametinib free base, or about 10 mg/m² based on mirdametinib free base.

[0108] In some embodiments of any of the methods described herein, the mirdametinib, or a pharmaceutically acceptable salt thereof, is administered twice daily in an amount of about 0.5 mg to about 10 mg based on mirdametinib free base, about 1 mg to about 9.5 mg based on mirdametinib free base, about 1.5 mg to about 9 mg based on mirdametinib free base, about 2 mg to about 8.5 mg based on mirdametinib free base, about 2.5 mg to about 8 mg based on mirdametinib free base, about 3 mg to about 7.5 mg based on mirdametinib free base, about 3.5 mg to about 7 mg based on mirdametinib free base, about 4 mg to about 6.5 mg based on mirdametinib free base, about 4.5 mg to about 6 mg based on mirdametinib free base, or about 5 mg to about 6 mg based on mirdametinib free base. In some embodiments of any of the methods described herein, the mirdametinib, or a pharmaceutically acceptable salt thereof, is administered twice daily in an amount of about 0.5 mg based on mirdametinib free base, about 1 mg based on mirdametinib free base, about 1.5 mg based on mirdametinib free base, about 2 mg based on mirdametinib free base, about 2.5 mg based on mirdametinib free base, about 3 mg based on mirdametinib free base, about 3.5 mg based on mirdametinib free base, about 4 mg based on mirdametinib free base, about 4.5 mg based on mirdametinib free base, about 5 mg based on mirdametinib free base, about 5.5 mg based on mirdametinib free base, about 6 mg based on mirdametinib free base, about 6.5 mg based on mirdametinib free base, about 7 mg based on mirdametinib free base, about 7.5 mg based on mirdametinib free base, about 8 mg based on mirdametinib free base, about 8.5 mg based on mirdametinib free base, about 9 mg based on mirdametinib free base, about 9.5 mg based on mirdametinib free base, or about 10 mg based on mirdametinib free base.

[0109] In some embodiments of any of the methods described herein, the mirdametinib, or a pharmaceutically acceptable salt thereof, is administered in a total daily dose that does not exceed about 10 mg/m² based on mirdametinib free base, about 9.5 mg/m² based on mirdametinib free base, about 9 mg/m² based on mirdametinib free base, about 8.5 mg/m² based on mirdametinib free base, about 8 mg/m² based on mirdametinib free base, about 7.5 mg/m² based on mirdametinib free base, about 7 mg/m² based on mirdametinib free base, about 6.5 mg/m² based on mirdametinib free base, about 6 mg/m² based on mirdametinib free base, about 5.5 mg/m² based on mirdametinib free base, about 5 mg/m² based on mirdametinib free base, about 4.5 mg/m² based on mirdametinib free base, about 4 mg/m² based on mirdametinib free base, about 3.5 mg/m² based on mirdametinib free base, about 3 mg/m² based on mirdametinib free base, about 2.5 mg/m² based on mirdametinib free base, about 2 mg/m² based on mirdametinib free base, or about 1.5 mg/m² based on mirdametinib free base.

[0110] In some embodiments of any of the methods described herein, the mirdametinib, or a pharmaceutically acceptable salt thereof, is administered in a total daily dose that does not exceed about 10 mg based on mirdametinib free base, about 9.5 mg based on mirdametinib free base, about 9 mg based on mirdametinib free base, about 8.5 mg based on mirdametinib free base, about 8 mg based on mirdametinib free base, about 7.5 mg based on mirdametinib

free base, about 7 mg based on mirdametinib free base, about 6.5 mg based on mirdametinib free base, about 6 mg based on mirdametinib free base, about 5.5 mg based on mirdametinib free base, about 5 mg based on mirdametinib free base, about 4.5 mg based on mirdametinib free base, about 4 mg based on mirdametinib free base, about 3.5 mg based on mirdametinib free base, about 3 mg based on mirdametinib free base, about 2.5 mg based on mirdametinib free base, about 2 mg based on mirdametinib free base, or about 1.5 mg based on mirdametinib free base.

[0111] In some embodiments of any of the methods described herein, the mirdametinib, or a pharmaceutically acceptable salt thereof, is administered as mirdametinib free base.

[0112] In some embodiments of any of the methods described herein, a therapeutically effective amount of mirdametinib free base is administered.

[0113] In some embodiments of any of the methods described herein, the mirdametinib free base is administered in an amount of about 1 mg/m² to about 10 mg/m² per day, about 1.5 mg/m² to about 9.5 mg/m² per day, about 2 mg/m² to about 9 mg/m² per day, about 2.5 mg/m² to about 8.5 mg/m² per day, about 3 mg/m² to about 8 mg/m² per day, about 3.5 mg/m² to about 7.5 mg/m² per day, about 4 mg/m² to about 7 mg/m² per day, about 4.5 mg/m² to about 6.5 mg/m² per day, or about 5 mg/m² to about 6 mg/m² per day. In some embodiments of any of the methods described herein, the mirdametinib free base is administered in an amount of about 1 mg/m² per day, about 1.5 mg/m² per day, about 2 mg/m² per day, about 2.5 mg/m² per day, about 3 mg/m² per day, about 3.5 mg/m² per day, about 4 mg/m² per day, about 4.5 mg/m² per day, about 5 mg/m² per day, about 5.5 mg/m² per day, about 6 mg/m² per day, about 6.5 mg/m² per day, about 7 mg/m² per day, about 7.5 mg/m² per day, about 8 mg/m² per day, about 8.5 mg/m² per day, about 9 mg/m² per day, about 9.5 mg/m² per day, or about 10 mg/m² per day.

[0114] In some embodiments of any of the methods described herein, the mirdametinib free base is administered in an amount of about 1 mg to about 10 mg per day, about 1.5 mg to about 9.5 mg per day, about 2 mg to about 9 mg per day, about 2.5 mg to about 8.5 mg per day, about 3 mg to about 8 mg per day, about 3.5 mg to about 7.5 mg per day, about 4 mg to about 7 mg per day, about 4.5 mg to about 6.5 mg per day, or about 5 mg to about 6 mg per day. In some embodiments of any of the methods described herein, the mirdametinib free base is administered in an amount of about 1 mg per day, about 1.5 mg per day, about 2 mg per day, about 2.5 mg per day, about 3 mg per day, about 3.5 mg per day, about 4 mg per day, about 4.5 mg per day, about 5 mg per day, about 5.5 mg per day, about 6 mg per day, about 6.5 mg per day, about 7 mg per day, about 7.5 mg per day, about 8 mg per day, about 8.5 mg per day, about 9 mg per day, about 9.5 mg per day, or about 10 mg per day.

[0115] In some embodiments of any of the methods described herein, the mirdametinib free base is administered in a single dosage form comprising about 0.1 mg/m² to about 10 mg/m², about 0.5 mg/m² to about 9.5 mg/m², about 1 mg/m² to about 9 mg/m², about 1.5 mg/m² to about 8.5 mg/m², about 2 mg/m² to about 8 mg/m², about 2.5 mg/m² to about 7.5 mg/m², about 3 mg/m² to about 7 mg/m², about 3.5 mg/m² to about 6.5 mg/m², about 4 mg/m² to about 6 mg/m², or about 4.5 mg/m² to about 5.5 mg/m². In some embodiments of any of the methods described herein, the

mirdametininib free base is administered in a single dosage form comprising about 0.1 mg/m², about 0.2 mg/m², about 0.3 mg/m², about 0.4 mg/m², about 0.5 mg/m², about 1 mg/m², about 1.5 mg/m², about 2 mg/m², about 2.5 mg/m², about 3 mg/m², about 3.5 mg/m², about 4 mg/m², about 4.5 mg/m², about 5 mg/m², about 5.5 mg/m², about 6 mg/m², about 6.5 mg/m², about 7 mg/m², about 7.5 mg/m², about 8 mg/m², about 8.5 mg/m², about 9 mg/m², about 9.5 mg/m², or about 10 mg/m².

[0116] In some embodiments of any of the methods described herein, the mirdametininib free base is administered in a single dosage form comprising about 0.1 mg to about 10 mg, about 0.5 mg to about 9.5 mg, about 1 mg to about 9 mg, about 1.5 mg to about 8.5 mg, about 2 mg to about 8 mg, about 2.5 mg to about 7.5 mg, about 3 mg to about 7 mg, about 3.5 mg to about 6.5 mg, about 4 mg to about 6 mg, or about 4.5 mg to about 5.5 mg. In some embodiments of any of the methods described herein, the mirdametininib free base is administered in a single dosage form comprising about 0.1 mg, about 0.2 mg, about 0.3 mg, about 0.4 mg, about 0.5 mg, about 1 mg, about 1.5 mg, about 2 mg, about 2.5 mg, about 3 mg, about 3.5 mg, about 4 mg, about 4.5 mg, about 5 mg, about 5.5 mg, about 6 mg, about 6.5 mg, about 7 mg, about 7.5 mg, about 8 mg, about 8.5 mg, about 9 mg, about 9.5 mg, or about 10 mg.

[0117] In some embodiments of any of the methods described herein, the mirdametininib free base is administered one, two, three, or four times per day. In some aspects, the mirdametininib free base is administered once daily. In some embodiments of any of the methods described herein, the mirdametininib free base is administered twice daily.

[0118] In some embodiments of any of the methods described herein, the mirdametininib free base is administered twice daily in an amount of about 0.5 mg/m² to about 10 mg/m², about 1 mg/m² to about 9.5 mg/m², about 1.5 mg/m² to about 9 mg/m², about 2 mg/m² to about 8.5 mg/m², about 2.5 mg/m² to about 8 mg/m², about 3 mg/m² to about 7.5 mg/m², about 3.5 mg/m² to about 7 mg/m², about 4 mg/m² to about 6.5 mg/m², about 4.5 mg/m² to about 6 mg/m², or about 5 mg/m² to about 6 mg/m². In some embodiments of any of the methods described herein, the mirdametininib free base is administered twice daily in an amount of about 0.5 mg/m², about 1 mg/m², about 1.5 mg/m², about 2 mg/m², about 2.5 mg/m², about 3 mg/m², about 3.5 mg/m², about 4 mg/m², about 4.5 mg/m², about 5 mg/m², about 5.5 mg/m², about 6 mg/m², about 6.5 mg/m², about 7 mg/m², about 7.5 mg/m², about 8 mg/m², about 8.5 mg/m², about 9 mg/m², about 9.5 mg/m², or about 10 mg/m².

[0119] In some embodiments of any of the methods described herein, the mirdametininib free base is administered twice daily in an amount of about 0.5 mg to about 10 mg, about 1 mg to about 9.5 mg, about 1.5 mg to about 9 mg, about 2 mg to about 8.5 mg, about 2.5 mg to about 8 mg, about 3 mg to about 7.5 mg, about 3.5 mg to about 7 mg, about 4 mg to about 6.5 mg, about 4.5 mg to about 6 mg, or about 5 mg to about 6 mg. In some embodiments of any of the methods described herein, the mirdametininib free base is administered twice daily in an amount of about 0.5 mg, about 1 mg, about 1.5 mg, about 2 mg, about 2.5 mg, about 3 mg, about 3.5 mg, about 4 mg, about 4.5 mg, about 5 mg, about 5.5 mg, about 6 mg, about 6.5 mg, about 7 mg, about 7.5 mg, about 8 mg, about 8.5 mg, about 9 mg, about 9.5 mg, or about 10 mg.

[0120] In some embodiments of any of the methods described herein, the mirdametininib free base is administered in a total daily dose that does not exceed about 10 mg/m², about 9.5 mg/m², about 9 mg/m², about 8.5 mg/m², about 8 mg/m², about 7.5 mg/m², about 7 mg/m², about 6.5 mg/m², about 6 mg/m², about 5.5 mg/m², about 5 mg/m², about 4.5 mg/m², about 4 mg/m², about 3.5 mg/m², about 3 mg/m², about 2.5 mg/m², about 2 mg/m², or about 1.5 mg/m².

[0121] In some embodiments of any of the methods described herein, the mirdametininib free base is administered in a total daily dose that does not exceed about 10 mg, about 9.5 mg, about 9 mg, about 8.5 mg, about 8 mg, about 7.5 mg, about 7 mg, about 6.5 mg, about 6 mg, about 5.5 mg, about 5 mg, about 4.5 mg, about 4 mg, about 3.5 mg, about 3 mg, about 2.5 mg, about 2 mg, or about 1.5 mg.

[0122] In some embodiments of any of the methods described herein, the mirdametininib, or a pharmaceutically acceptable salt thereof, exhibits high blood-brain-barrier penetration.

[0123] In some embodiments of any of the methods described herein, the human has had no prior exposure to MEK inhibitors. In some embodiments of any of the methods described herein, the human has not responded to prior treatment to one or more MEK inhibitors.

[0124] In some embodiments of any of the methods described herein, the mirdametininib, or a pharmaceutically acceptable salt thereof, is administered orally. In some aspects, the mirdametininib, or a pharmaceutically acceptable salt thereof, is administered orally as a solid dosage form. In some embodiments of any of the methods described herein, the solid dosage form is a tablet or capsule. In some embodiments of any of the methods described herein, the solid dosage form is a capsule. In some aspects, the mirdametininib, or a pharmaceutically acceptable salt thereof, is dispersible in a potable liquid or orodispersible in a patient's saliva. In one embodiment, the mirdametininib, or a pharmaceutically acceptable salt thereof, is administered as a dispersible formulation (such as a 0.5 mg or 1 mg mirdametininib dispersible tablet) as described in U.S. Pat. No. 11,571,402, which is hereby incorporated by reference.

[0125] In some embodiments of any of the methods described herein, the mirdametininib, or a pharmaceutically acceptable salt thereof, is administered as a monotherapy to treat the neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas.

[0126] In some embodiments of any of the methods described herein, the mirdametininib, or a pharmaceutically acceptable salt thereof, is administered in combination with another active ingredient and/or surgery to treat the neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas.

EXAMPLES

Example 1: Phase I/II Evaluation of Single Agent Mirdametininib, a Brain-Penetrant MEK1/2 Inhibitor, for the Treatment of Children, Adolescents, and Young Adults with Brain Cancer

[0127] A multi-arm phase I/II trial of mirdametininib is to be conducted in patients ≥ 2 and < 25 years with brain cancer. Phase I requires participants have no prior exposure to MEK inhibitors and recurrent/progressive disease with biopsy-proven evidence of MAPK pathway activation. Three esca-

lating dose levels (2 mg/m²/dose BID, 2.5 mg/m²/dose BID and 3 mg/m²/dose BID) are planned using a rolling 6 design.

[0128] The median age enrolled in the study is 10 years old (3-21). No dose-limiting toxicities occur for all three dose levels. No MEK related retinopathy or cardiopathy is observed. No disease progressions are occurred. Thus far, mirdametinib is well-tolerated and clinically promising when dosed continuously in patients with brain cancer.

Example 2: Phase 2b Trial of Mirdametinib in
Adult and Pediatric Patients with Neurofibromatosis
Type 1 (NF1)-Associated Inoperable Plexiform
Neurofibromas (PNs) that are Progressing or
Causing Significant Morbidity

[0129] This study is to evaluate the efficacy, safety, and tolerability of mirdametinib in participants ≥ 2 years of age with an inoperable NF1-associated plexiform neurofibromas (PNs) that is progressing and/or causing significant morbidity.

[0130] Approximately 120 participants will be screened (assessed for eligibility as described below) to achieve approximately 100 participants assigned to study treatment. Of these participants, approximately 50 will be ≥ 18 years of age and approximately 50 will be 2 to 17 years of age.

[0131] Participants will be screened for up to 28 days prior to the first dose of study treatment mirdametinib. Study treatment will be administered orally at a twice daily (BID) dose specified in the table below. Dosing will be on a 28-day Cycle (4-week course) with a 3 week on/1 week off schedule. The treatment period will last for up to 24 Cycles followed by a 30-day Safety Follow-Up period.

Subject's BSA (m ²)	≤ 0.69	0.7 to 1.04	1.05 to 1.49	≥ 1.5
BID dose (mg)	1	2	3	4

[0132] A partial response is defined as PN decrease $\geq 20\%$ compared to baseline using centrally read MRI volumetric analysis.

Inclusion Criteria

[0133] Patients are eligible to be included in the study only if all of the following criteria apply:

[0134] 1. Participant must be ≥ 2 years of age inclusive, at the time of signing the informed consent/assent.

[0135] 2. Participants must have either the clinical diagnosis of NF1 using the National Institute of Health (NIH) Consensus Conference criteria of at least 1 other diagnostic criterion (Inclusion 2.1-2.6, see below) in addition to the presence of PN, or have a constitutional NF1 mutation documented in a Clinical Laboratory Improvement Amendments/College of American Pathologists certified lab; additional criteria are as follows:

[0136] 2.1. Six or more café-au-lait macules with a diameter >5 mm in prepubertal and >15 mm in post-pubertal individuals, respectively

[0137] 2.2. Freckling in axilla or inguinal regions;

[0138] 2.3. Optic glioma;

[0139] 2.4. Two or more Lisch nodules;

[0140] 2.5. A distinctive bony lesion (dysplasia of the sphenoid bone or dysplasia of thinning of long bone cortex);

[0141] 2.6. A first degree relative with NF1.

[0142] 3. Participants must have PN that is progressive (Inclusion 3.1) OR causing significant morbidity, such as (but not limited to) head and neck lesions that are compromising the airway or great vessels, brachial or lumbar plexus lesions that are causing nerve compression and loss of function, lesions causing major deformity or are significantly disfiguring (Inclusion 3.2), lesions of the extremity that cause limb hypertrophy or loss of function, and painful lesions. Participants with paraspinal PNs will be eligible for this study. Histologic confirmation of tumor is not necessary in the presence of consistent clinical and radiographic findings but should be considered if malignant degeneration of a PN is clinically suspected;

[0143] 3.1. For participants enrolled for tumor progression, progression is defined as:

[0144] 3.1.1. A measurable increase in PN size ($\geq 20\%$ increase in volume) documented by comparison of two MRI scans in the time period of 12 months or less prior to first dose of study treatment (mirdametinib).

[0145] 3.2. For participants enrolled for a "major deformity" or "significantly disfiguring" tumor, eligible tumors will be limited to tumors of the head and neck or those on other areas of the body that are unable to be concealed by standard garments.

[0146] 4. Participant has a PN that is deemed inoperable, defined as a PN that cannot be completely surgically removed without risk for substantial morbidity due to: encasement of or close proximity to vital structures, invasiveness, or high vascularity of the PN, or the participant refuses surgery. Participants who previously underwent surgery for a PN will be eligible to enter the study after the surgery, provided the PN was incompletely resected and is evaluable by volumetric analysis.

[0147] 5. Participants must have a target PN, defined as the clinically most relevant PN, amenable to volumetric MRI analysis. For the purpose of this study, the target PN must be seen on at least 3 consecutive MRI slices and the field of view must contain the entire tumor of interest. As determined by central radiologic review, a target PN must be analyzable by volumetrics, at least 5 mL in volume, and will be classified as "typical PN", "nodular PN", or "solitary nodular PN" prior to first dose of study treatment.

[0148] 6. Participants ≥ 18 years of age must have a PN amenable to a percutaneous biopsy and must be willing to undergo pre-, and on treatment tumor biopsies providing fresh tumor tissue; there should be no contraindication for serial biopsy; Patients 2 to 17 years of age should not undergo biopsy unless there is a clinical indication to obtain fresh tumor tissue.

[0149] 7. Participants ≥ 16 years of age must have a Karnofsky performance level of $\geq 60\%$, and participants < 16 years must have a Lansky performance of $\geq 60\%$.

[0150] 8. Participant has adequate organ and bone marrow function as defined by the following screening laboratory values:

[0151] 8.1. Absolute neutrophil count ≥ 1500 cells/ μ L;

[0152] 8.2. Platelets $\geq 100 \times 10^3/\mu$ L;

[0153] 8.3. Hemoglobin ≥ 9.5 g/dL;

- [0154] 8.4. Serum albumin ≥ 2.8 g/dL;
- [0155] 8.5. Calculated creatinine clearance at Screening >60 mL/min (by Cockcroft-Gault formula) OR a normal serum creatinine based on age described in the table below.

Age (years)	Maximum Serum Creatinine (mg/dL)
≤ 5	0.8
>5 and ≤ 10	1.0
>10 and ≤ 15	1.2
>15	1.5

- [0156] 9. Participant has the ability to swallow capsules whole.
- [0157] 10. Participant is willing and able to comply with all aspects of the protocol.
- [0158] 11. Participant must weigh at least 10 kg, inclusive, at the time of signing the informed consent/assent.
- [0159] 12. Participant must have a body surface area (BSA) of at least 0.4 m^2 (inclusive) calculated using the Du Bois formula ($\text{BSA} = 0.007184 \times W^{0.425} \times H^{0.725}$).
- [0160] 13. Male or Female
- [0161] Contraceptive use by men or women should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies.
- [0162] a. Male participants:
- [0163] Male participants are eligible to participate if they agree to the following during the treatment period and for at least 90 days after the last dose of study treatment:
- [0164] Refrain from donating sperm
PLUS either:
- [0165] Be abstinent from heterosexual intercourse as their preferred and usual lifestyle (abstinent on a long term and persistent basis) and agree to remain abstinent OR
- [0166] Must agree to use a male condom when having sexual intercourse with a woman of child bearing potential (WOCBP).
- [0167] b. Female participants:
- [0168] Female participants are eligible to participate if they are not pregnant or breastfeeding, and at least one of the following conditions applies:
- [0169] Is not a woman of childbearing potential
OR
- [0170] Is a WOCBP and using a contraceptive method that is highly effective (with a failure rate of $<1\%$ per year), preferably with low user dependency, during the treatment period and for at least 30 days after the last dose of study treatment and agrees not to donate eggs (ova, oocytes) for the purpose of reproduction during the study and for a period of 90 days.
- A WOCBP must have a negative serum pregnancy test result at Screening and a negative urine pregnancy test result at the Baseline visit prior to the first dose of study treatment.

Exclusion Criteria

- [0171] Participants are excluded from the study if any of the following criteria apply:
- [0172] 1. Participant has a Screening alanine transaminase (ALT) value of $>2.0 \times$ upper limit of normal (ULN);
- [0173] 2. Participant has a Screening total bilirubin value of $>1.5 \times \text{ULN}$ (isolated bilirubin $>1.5 \times \text{ULN}$ is acceptable if bilirubin is fractionated and direct bilirubin $<35\%$);
- [0174] 3. Participant has a history of malignancy associated hypercalcemia;
- [0175] 4. Participant has an active parathyroid disorder, hyperphosphatemia at Screening (serum phosphorus $>1 \times \text{ULN}$), and/or serum calcium (mg/dL) $\times$ serum phosphorus (mg/dL) product >70 at Screening.
- [0176] 5. Any clinically significant active or known history of liver disease, or known hepatic or biliary abnormalities (with the exception of Gilbert's syndrome or asymptomatic gallstones);
- [0177] 5.1 Hepatitis serology and viral load will be tested at Screening. Patients who are hepatitis B surface antigen (HBsAg) positive or hepatitis C virus (HCV) antibody positive at Screening must not be enrolled until further definite testing with hepatitis B virus (HBV) deoxyribonucleic acid (DNA) titers is <500 IU/mL or HCV ribonucleic acid (RNA) polymerase chain reaction test is negative;
- [0178] 6. Lymphoma, leukemia, or any malignancy (including malignant glioma or malignant peripheral nerve sheath tumor (MPNST)) within the past 5 years except for basal cell or squamous epithelial carcinomas of the skin that have been resected with no evidence of metastatic disease for 3 years;
- [0179] 7. Breast cancer within the past 10 years;
- [0180] 8. Participants with evidence of an active optic glioma or other low-grade glioma, requiring treatment with chemotherapy or radiation therapy. Participants not requiring treatment are eligible. Ophthalmological findings secondary to long-standing optic pathway glioma (such as visual loss, optic nerve pallor or strabismus) or long-standing orbito-temporal PN (such as visual loss, strabismus) will not be considered a significant abnormality for the purposes of the study;
- [0181] 9. Participant has abnormal QT interval corrected by Fridericia's formula (>450 msec for male participants, >470 msec for female participants, or >480 msec for participants with bundle branch block) after electrolytes have been corrected (triplicate ECG readings taken 2 to 3 minutes apart and averaged) at Screening;
- [0182] 10. Participant has experienced any of the following within 6 months (24 weeks) of signing informed consent/assent: clinically significant cardiac disease, myocardial infarction, severe/unstable angina, coronary/peripheral artery bypass graft, cerebrovascular accident, transient ischemic attack, or symptomatic pulmonary embolism;
- [0183] 11. Participant has ever had a recorded left ventricular ejection fraction (LVEF) $<55\%$ as assessed by echocardiogram, OR has a history of congestive heart failure;
- [0184] 12. Participant has a history of, or evidence of, retinal pathology on ophthalmologic examination that is considered a risk factor for central serous retinopathy, retinal vein occlusion (RVO), or neovascular macular degeneration. Participants will be excluded

from study participation if they currently are known to have any of the following risk factors for RVO:

- [0185] 12.1 Intraocular pressure ≥ 21 mmHg;
- [0186] 12.2 Serum cholesterol >300 mg/dL;
- [0187] 12.3 Serum triglycerides >300 mg/dL;
- [0188] 12.4 Hyperglycemia (fasting blood glucose >125 mg/dL or random blood glucose >200 mg/dL);
- [0189] 12.5 Age specific hypertension
 - [0190] i. Participants ≥ 13 years of age with a blood pressure $\geq 140/90$ mm Hg
 - [0191] ii. Participants ≤ 12 years of age with a blood pressure $>95^{th}$ percentile for age+12 mmHg (Section Error! Reference source not found.);
- [0192] 13. Participant has a history of glaucoma;
- [0193] 14. Participant has a history of a positive human immunodeficiency virus (HIV) antibody test;
- [0194] 15. Participant has a known malabsorption syndrome or preexisting gastrointestinal conditions that may impair absorption of mirdametininib (e.g., gastric bypass, lap band, or other gastric procedures). Delivery of mirdametininib via nasogastric tube or gastrostomy tube is not allowed.
- [0195] 16. Participant has received NF1 PN-targeted therapy (e.g., MEK inhibitors, farnesyltransferase inhibitors, kinase inhibitors, etc.) within 28 days of first dose of study treatment (or 5.5 half-lives, whichever is longer). If participant enrolls with progression and no associated morbidities, NF1-targeted therapy must not be administered after the observed progression (Inclusion Criterion 3.1.1). All toxicities from prior therapy must resolve to $\leq$ Grade 1 or Baseline;
- [0196] 17. Participant previously received or is currently receiving therapy with mirdametininib;
- [0197] 18. Participant is receiving systemic or ocular glucocorticoid therapy (with the exception of participants with endocrine deficiencies who are allowed to receive physiologic or stress doses of steroids, if necessary) within 14 days prior to first dose of study treatment;
- [0198] 19. Participant has received radiation therapy within the 6 months prior to signing of informed consent/assent. Participants who have received radiation to the orbit at any time are excluded;
- [0199] 20. Current enrollment or past participation in any other clinical study (excluding observational studies) within 28 days of signing of informed consent/assent;
- [0200] 21. Participant is unable to tolerate MRI or for whom MRI is contraindicated;
- [0201] 22. Tumor is not able to be reliably evaluated by MRI volumetric analysis;
- [0202] 23. Sensitivity to the study treatment, or components thereof, or drug or other allergy that, in the opinion of the investigator or medical monitor, contraindicates participation in the study;
- [0203] 24. Participant with active bacterial, fungal, or viral infection including but not limited to the use of antibiotics, antifungals, or antiviral agents at the time of screening;
- [0204] 25. Underlying medical conditions, laboratory abnormality, or alcohol or drug abuse or dependence that, in the investigator's opinion, will be unfavorable for the administration of study treatment or affect the explanation of drug toxicity or adverse events; or

insufficient compliance during the study according to investigator's judgement; or

- [0205] 26. Participant has experienced other severe acute or chronic medical or psychiatric conditions, including recent (within 1 year of signing informed consent/assent) or active suicidal ideation or behavior, or a laboratory abnormality that may increase the risk associated with study participation or study treatment administration or may interfere with the interpretation of study results and, in the judgment of the Investigator, would make the participant inappropriate for entry into this study.

Supportive Care

Dermatologic Adverse Events:

[0206] The use of medications for the supportive care of rash is permitted. Early initiation of treatment for rashes is strongly recommended to minimize the duration and severity of the adverse event.

[0207] Acneiform rash: Pustular rash may be treated with topical clindamycin gel or lotion applied BID. In severe cases, semisynthetic oral tetracyclines such as doxycycline or minocycline may also be useful for older children, adolescents, and adults, but should be avoided in children younger than 8 years old because of risk to tooth development.

[0208] Eczematous rash/xerosis: Eczematous/dry skin rash and other macular (non-acneiform) rash should be treated with a moisturizer such as Cerave or Eucerin or another equivalent product. A low potency topical steroid such as betamethasone valerate lotion (0.05%), desonide cream (0.05%), fluocinonide acetone solution (0.01%), dexamethasone sodium phosphate cream (0.1%), hydrocortisone acetate cream (1%), methylprednisolone acetate cream (0.25%) or equivalent may also be used if symptomatic.

[0209] Ketoconazole shampoo should be used for any rash involving the scalp.

[0210] Paronychia: Paronychia if acute and non-surgical (i.e., no fluctuance suggesting an abscess) can resolve with warm soaks only applied 3 to 4 times daily. If there is extensive redness suggesting cellulitis, OR if there is non-surgical paronychia but the participant is a diabetic or is immuno-compromised, then an oral antibiotic that covers *Staphylococcus aureus* should be started. The choice of antibiotics includes a *Staphylococcus aureus* covering penicillin/clindamycin/first generation cephalosporin/Augmentin (amoxicillin and clavulanate).

[0211] If an abscess develops, surgical treatment with incision and drainage with or without debridement should be done. Any infectious organisms identified should be treated accordingly. If the participant has diabetes or is immune compromised, oral antibiotics ensuring coverage for *Staphylococcus aureus* (see above) should be started prior to a culture and sensitivity report. Once culture report is obtained, the antibiotic therapy should be adjusted as appropriate.

Prohibited or Restricted Concomitant Medications/Treatments

[0212] Prior use of mirdametininib is prohibited.

[0213] Alternative therapy for the treatment of PNs (e.g. MEK inhibitors, farnesyltransferase inhibitors, kinase inhibitors, etc.) within 28 days (or 5.5 half-lives, whichever is longer) of first dose of study treatment and throughout the treatment period is prohibited. If participant enrolls with progression and no associated morbidities, NF1-targeted therapy must not be administered after the observed progression (Inclusion Criterion 3.1.1)

[0214] Medical treatment (e.g. chemotherapy, biologic therapy, radiation therapy) directed towards any NF1-related tumor such as optic pathway glioma is prohibited throughout the treatment period.

[0215] The use of chronic systemic or ocular glucocorticoid therapy is prohibited within the 14 days prior to first dose of study treatment and throughout the treatment period (with the exception of participants with endocrine deficiencies who are allowed to receive physiologic or stress doses of steroids, if necessary). In addition, corticosteroids are permissible as premedication for blood product transfusions, or as treatment for an acute allergic reaction or bronchospasm.

[0216] All publications, patents, and patent applications mentioned in this specification are incorporated herein by reference in their entirety to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference in its entirety. Where a term in the present application is found to be defined differently in a document incorporated herein by reference, the definition provided herein is to serve as the definition for the term.

[0217] While the invention has been described in connection with specific aspects thereof, it will be understood that invention is capable of further modifications and this application is intended to cover any variations, uses, or adaptations following, in general, the principles and including such departures from the present disclosure that come within known or customary practice within the art to which the invention pertains and can be applied to the essential features hereinbefore set forth, and follows in the scope of the claimed.

1. (canceled)

2. The method of claim 25, wherein the patient has symptomatic, inoperable plexiform neurofibromas.

3. The method of claim 25, wherein the patient has neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) that is progressing or causing significant morbidity.

4. The method of claim 25, wherein the patient has progressive PN.

5. The method of claim 25, wherein the patient has PNs that cause significant morbidity.

6. The method of claim 25, wherein the patient has head and neck lesions that are compromising the airway or great vessels, brachial or lumbar plexus lesions that are causing nerve compression and loss of function, lesions causing major deformity or are significantly disfiguring, lesions of the extremity that cause limb hypertrophy or loss of function, or painful lesions.

7. The method of claim 6, wherein the lesions causing major deformity or are significantly disfiguring are tumors

of the head and neck or those on other areas of the body that are unable to be concealed by standard garments.

8. The method of claim 25, wherein the patient has paraspinal lesions.

9. The method of claim 25, wherein the patient has a Lansky performance of at least 60%.

10. The method of claim 25, wherein the patient has the clinical diagnosis of NF1 using the NIH Consensus Conference and one or more of the following:

- (a) six or more café-au-lait macules with a diameter >5 mm in prepubertal and >15 mm in post-pubertal individuals;
- (b) freckling in axilla or inguinal regions;
- (c) optic glioma;
- (d) two or more Lisch nodules;
- (e) a distinctive bony lesion (dysplasia of the sphenoid bone or dysplasia of thinning of long bone cortex); and
- (f) a first degree relative with NF1.

11. The method of claim 25, wherein the patient has a constitutional NF1 mutation documented in a Clinical Laboratory Improvement Amendments/College of American Pathologists certified lab.

12. The method of claim 25, wherein the patient either (a) has a parent diagnosed with NF1 and one or more criteria of (1) through (7) or (b) does not have a parent diagnosed with NF1 but has two or more criteria of (1) through (7):

- (1) six or more café-au-lait macules over 5 mm in greatest diameter in prepubertal individuals and over 15 mm in greatest diameter in post-pubertal individuals;
- (2) freckling in the axillary or inguinal region;
- (3) two or more neurofibromas of any type or one plexiform neurofibroma;
- (4) optic pathway glioma;
- (5) two or more iris Lisch nodules identified by slit lamp examination or two or more choroidal abnormalities (defined as bright, patchy nodules imaged by optical coherence tomography (OCT)/near-infrared reflectance (NIR) imaging;
- (6) a distinctive osseous lesion (such as sphenoid dysplasia, anterolateral bowing of the tibia, or pseudarthrosis of a long bone); and
- (7) a heterozygous pathogenic NF1 variant with a variant allele fraction of 50% in apparently normal tissue such as white blood cells.

13-15. (canceled)

16. The method of claim 25, wherein over each four week period, the mirdametininib is administered for the first three weeks and discontinued for the last one week.

17. The method of claim 25, wherein the patient has at least a 20% reduction in plexiform neurofibroma volume as determined by volumetric magnetic resonance imaging analysis following treatment with mirdametininib.

18. The method of claim 25, wherein the treatment results in decreased pain intensity.

19. The method of claim 25, wherein the treatment results in decreased pain interference.

20. (canceled)

21. The method of claim 26, wherein the adverse event resulting in the dose reduction is acneiform.

22. The method of claim 25, wherein the method further comprises prior to treatment (i) determining whether to select mirdametininib as a treatment for the patient, and (ii) selecting mirdametininib as a treatment for the patient at least partially based on its objective response rate, where the

objective response rate is defined as at least a 20% decrease in tumor size using centrally read MRI volumetric analysis.

23. The method of claim **22**, wherein in step (i), mirdametinib is selected based on a response rate of at least 70%.

24. (canceled)

25. A method of treating a pediatric human patient 2 to 15 years of age who has neurofibromatosis type 1 (NF1) associated inoperable plexiform neurofibromas (PN) and a body surface area of 0.7 to 1.04 m², the method comprising orally administering to the patient 2 mg mirdametinib twice daily.

26. The method of claim **25**, wherein upon an adverse event, the dose is reduced to 2 mg mirdametinib administered in the morning and 1 mg mirdametinib administered in the evening.

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